



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

13-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	C	A	B	D	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A	A	B,C	A,D	A,C,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.00	162.00	4.00	3.46	500.00	2.50

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	A	B	B	C	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	A,C	A,B,D	B,C,D	A,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	40.00	53.25	3.00	5.00	4.00	20.00

PART-3 : MATHEMATICS

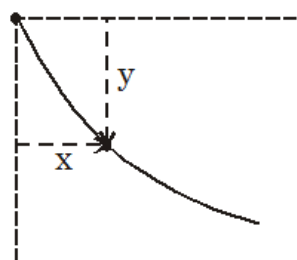
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	B	C	C	C	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	B,D	A,C	A,C,D	A,C,D	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	48.00	1.50	1.00	256.00	3.24	9.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)



$$y = \frac{1}{2}gt^2$$

$$y = \frac{1}{2} \times 10 \times t^2$$

$$y = 5t^2$$

time to hit the ground

$$\frac{1}{2}gt^2 = H = s_y = 80$$

$$\Rightarrow t = 4 \text{ s}$$

Now, $v_x = ay$

$$\frac{dx}{dt} = ay$$

$$\frac{dx}{dt} = a(5t^2)$$

$$\int_{x_1}^{x_2} dx = \frac{3 \times 5}{8} \int_0^4 t^2 dt$$

$$\Delta x = \frac{3 \times 5}{8} \times \frac{4^3}{3}$$

$$\Delta x = 40 \text{ m Ans.}$$

2. Ans (C)

$$v_{\text{avg}} = \frac{\text{total displacement}}{\text{total time}} = \frac{x_2 - x_1}{\Delta T}$$

$$x_1 \big|_{(t=0)} = A \sin(\omega t + \theta) = A \sin \theta$$

$$x_2 \Big|_{t=\frac{T}{4}} = A \sin\left(\frac{\omega T}{4} + \theta\right) = A \cos \theta$$

$$v_{\text{avg}} = \frac{A (\cos \theta - \sin \theta)}{\frac{T}{4}}$$
$$= \frac{2A\omega (\cos \theta - \sin \theta)}{\pi}$$

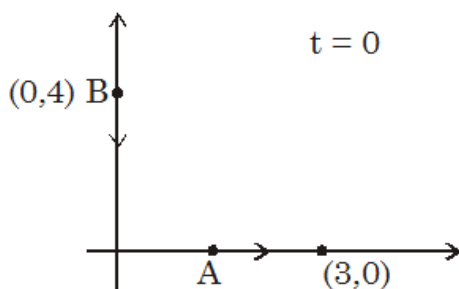
Or use

$$\langle f(t) \rangle = \frac{\int_{T_1}^{T_2} f(t) dt}{T_2 - T_1}$$

to get the average value of velocity where

$$f(t) = \frac{dx}{dt} = A\omega \cos(\omega t + \theta)$$

3. Ans (A)



$$x = 3 - 2 \cos \omega t$$

$$y = 4 - 2 \sin \omega t$$

$$d^2 = x^2 + y^2$$

$$d^2 = 29 - 12 \cos \omega t - 16 \sin \omega t$$

$$\frac{d}{dt}(d^2) = 0 \text{ Maxima/Minima}$$

$$\Rightarrow 12 \sin \omega t - 16 \cos \omega t = 0$$

$$\Rightarrow \tan \omega t = \frac{4}{3}$$

$$\omega t = 53^\circ \qquad d_{1(\min)} = 3\text{cm}$$

$$\omega t = 180^\circ + 53^\circ \quad d_{2(\max)} = 7\text{cm}$$

4. Ans (B)

$$\mu \sin \theta_c = 1 \times \sin 90^\circ$$

$$\sin \theta_c = \frac{1}{\mu} \quad \dots \text{(i)}$$

$$\theta_c + r = 90^\circ$$

$$r = (90^\circ - \theta_c)$$

Snell's law at top interface

$$1 \times \sin i = \mu \sin r$$

$$\sin i = \mu \sin (90 - \theta_c)$$

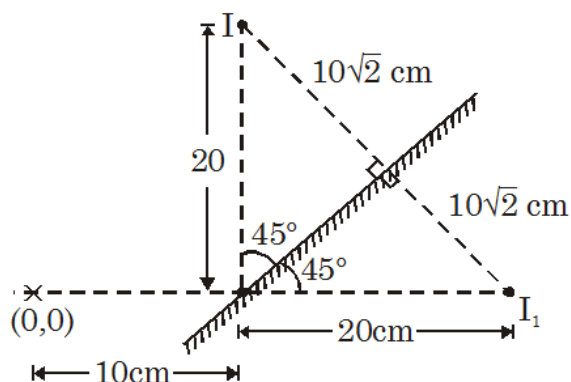
$$\sin i = \mu \cos \theta_c$$

$$\sin^2 i = \mu^2 \left(1 - \frac{1}{\mu^2} \right)$$

$$\mu = \sqrt{1 + \sin^2 i}$$

5. Ans (D)

Image due to lens is at 30 cm on right i.e. 20 cm inside mirror



Final image I at (10, 20) cm.

6. Ans (B)

Equation of trajectory

$$y = x \tan \theta - \frac{gx^2}{2u^2} \sec^2 \theta$$

(α, β) should satisfy the above equation if projectile is going to hit the target.

$$\beta = \alpha \tan \theta - \frac{g\alpha^2}{2(2g\alpha)}(1 + \tan^2 \theta)$$

$$\beta = \alpha \tan \theta - \frac{\alpha}{4} - \frac{\alpha}{4} \tan^2 \theta$$

$$\left(\frac{\alpha}{4}\right) \tan^2 \theta - (\alpha) \tan \theta + \left(\beta + \frac{\alpha}{4}\right) = 0$$

If discriminant of above quadratic equation in $\tan^2\theta$ is negative, there can be no real solution and thus it will become impossible to hit the target.

$$\text{i.e. } \alpha^2 - 4 \left(\frac{\alpha}{4} \right) \left(\beta + \frac{\alpha}{4} \right) < 0$$

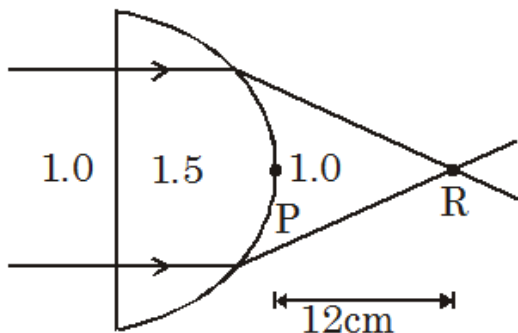
$$\Rightarrow 4\beta > 3\alpha$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$



$$\frac{1}{v} - \frac{1.5}{-\infty} = \frac{1 - 1.5}{-6} \Rightarrow v = 12\text{cm}$$

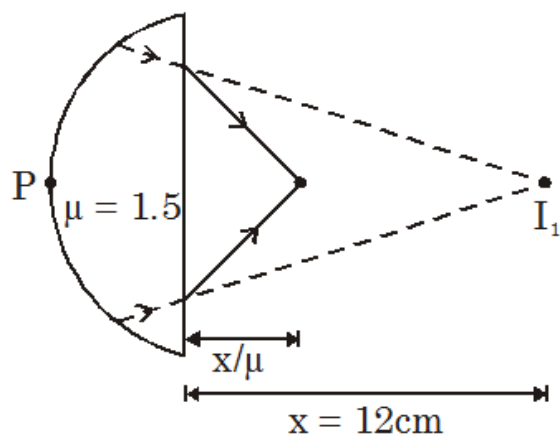
For figure (ii)

There will be refraction twice.

First image by curved surface.

$$\frac{1.5}{v} - \frac{1}{-\infty} = \frac{1.5 - 1}{+6}$$

$$\Rightarrow \frac{3}{2v} = \frac{1}{12} \Rightarrow v = 18\text{cm from P.}$$



$$\therefore \frac{x}{\mu} = \frac{12}{3/2} = 8\text{cm}$$

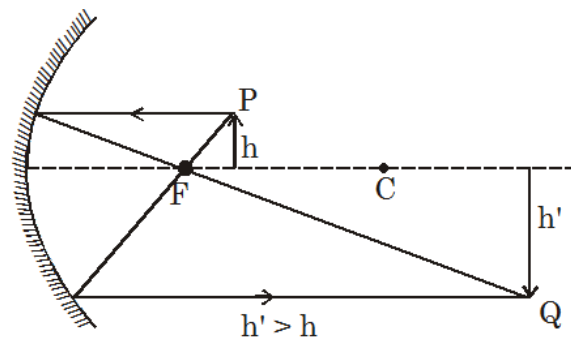
From P = 6 + 8 = 14 cm

8. **Ans (A)**

Only real image can be formed as image will be inverted w.r.t. object. So both P and Q should be on same side of mirror $\Rightarrow$ converging mirror.

Moreover, distance of P from xy is smaller than distance of Q from xy.

Hence, P will be closer to the mirror than Q.



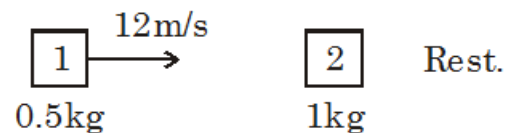
9. **Ans (B,C)**

$$T = 2\pi\sqrt{\frac{m_2}{k}}$$

$$\Rightarrow 0.628 = 6.28\sqrt{\frac{m_2}{100}}$$

$$\Rightarrow m_2 = 1\text{ kg}$$

Collision of m_1 & m_2



After collision



$\therefore$ by momentum conservation

$$1 \times v_2 - \frac{1}{2} \times v_1 = \frac{1}{2} \times 12$$

$$2v_2 - v_1 = 12 \quad \dots (i)$$

For elastic collision

$$\frac{v_1 + v_2}{12} = e = 1$$

$$\Rightarrow v_2 + v_1 = 12 \quad \dots (ii)$$

$$v_2 = 8\text{ m/s} \quad v_1 = 4\text{ m/s}$$

Time to fall off the cliff

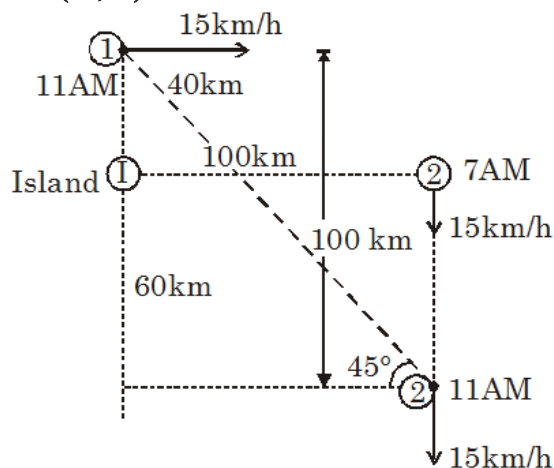
$$\frac{1}{2}gt^2 = 4.9 \Rightarrow t = 1\text{ s}$$

$$\therefore d = 4\text{ m}$$

Speed upon hitting the ground

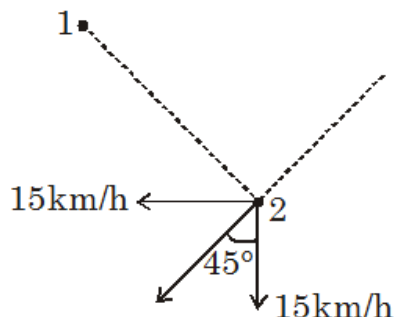
$$\sqrt{v_{1x}^2 + v_{1y}^2} > 4\text{ m/s.}$$

10. Ans (A,D)



At 11 AM both have min separation.

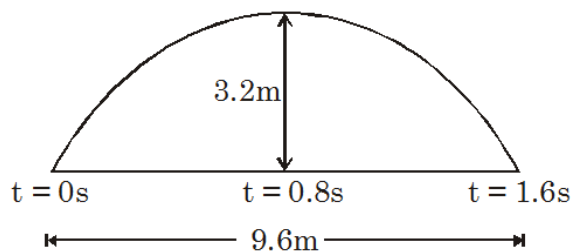
In frame of ship (1)



11. Ans (A,C,D)

$$v_x = 6 \text{ m/s}, v_y = 8 \text{ m/s}, t = 0$$

Uninterrupted parabolic path of the object.



12. Ans (B,C)

$$x = a \sin^2 \theta$$

$$x = a \sin^2 \left(\omega t - \frac{\pi}{4} \right)$$

$$x = \frac{a}{2} - \frac{a}{2} \cos \left(2\omega t - \frac{\pi}{2} \right)$$

$$v = \frac{dx}{dt} = +\frac{a}{2} (2\omega) \sin 2\theta$$

$$\text{where } \theta = \left(\omega t - \frac{\pi}{4} \right)$$

$$v = 2a\omega \sin \theta \cos \theta$$

$$v^2 = 4a^2 \omega^2 \sin^2 \theta \cos^2 \theta$$

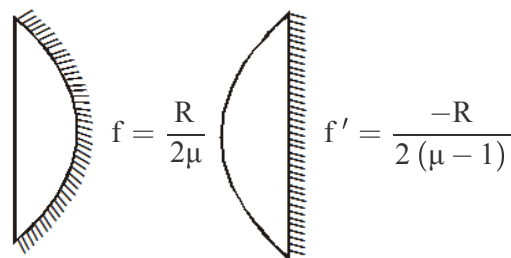
$$v^2 = 4\omega^2 (a \sin^2 \theta) a (1 - \sin^2 \theta)$$

$$v^2 = 4\omega^2 x (a - x)$$

PART-1 : PHYSICS

SECTION-II

1. Ans (3.00)



$$\text{Here } \left| \frac{f'}{f} \right| = \frac{30}{10} = 3$$

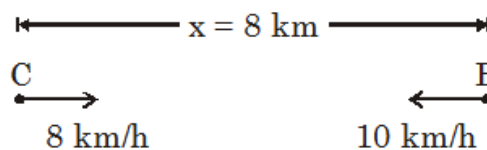
$$\Rightarrow \frac{R}{2(\mu-1)} \times \frac{2\mu}{R} = 3 \Rightarrow \mu = \frac{3}{2}$$

$$\text{According to the question } \mu = \frac{P}{2}$$

$$\therefore P = 3 \text{ Ans.}$$

2. Ans (162.00)

Situation when the fly is at B of the first time



Time taken by fly to meet car and reach B for

$$2^{\text{nd}} \text{ time is } = 2 \times \frac{x}{8+10} = \frac{x}{9}$$

New separation between car and B when fly reaches B

For the 2nd time is

$$= x - 8 \times \frac{x}{9} = \frac{x}{9}$$

Separation between car and B when the fly

$$\text{reaches B for } n^{\text{th}} \text{ time} = \frac{x}{9^{n-1}}$$

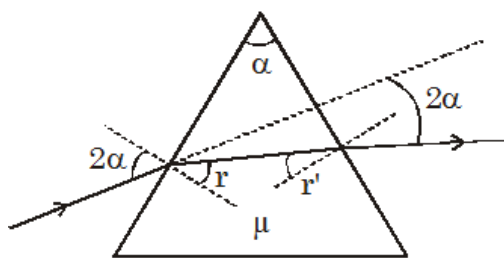
$\therefore$ When fly reaches B for 3rd time separation

$$\text{between car and point B is } \frac{x}{81}$$

where $x = 8 \text{ km}$

$$\therefore \text{Ans.} = \frac{8}{81} = \frac{16}{162}$$

3. Ans (4.00)



$$r + r' = \alpha$$

$$r' = \alpha - r \quad \dots (i)$$

$$i + e = \delta + A$$

$$\Rightarrow 2\alpha + e = 2\alpha + \alpha \Rightarrow e = \alpha$$

Snell's law at point of incidence

$$1 \times \sin 2\alpha = \mu \sin r \quad \dots (ii)$$

At the point of emergence

$$\mu \sin (\alpha - r) = \sin \alpha \quad \dots (iii)$$

$$\frac{\sin \alpha}{\mu} = \sin \alpha \cos r - \cos \alpha \sin r$$

$$\frac{\sin \alpha}{\mu} = \sin \alpha \cos r - \frac{\cos \alpha (\sin 2\alpha)}{\mu}$$

$$\Rightarrow 2\cos^2 \alpha + 1 = \mu \cos r \quad \dots (iv)$$

From (ii)

$$\sin r = \frac{\sin 2\alpha}{\mu}$$

$$1 - \cos^2 r = \frac{\sin^2 2\alpha}{\mu^2}$$

$$\mu \cos r = \sqrt{\mu^2 - 1 + \cos^2 2\alpha} \quad \dots (v)$$

From (iv) and (v)

$$(2\cos^2 \alpha + 1)^2 = (\mu - 1 + \cos^2 2\alpha)$$

$$(\cos 2\alpha + 2)^2 = \mu^2 - 1 + \cos^2 2\alpha$$

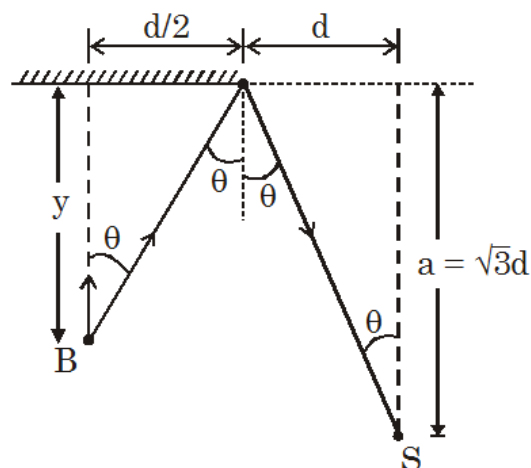
$$\Rightarrow 4 + 4 \cos 2\alpha = \mu^2 - 1$$

$$\Rightarrow 8 \cos^2 \alpha = \mu^2 - 1$$

$$\Rightarrow \cos^2 \alpha = \frac{\mu^2 - 1}{8} = \frac{\mu^2 - 1}{2x}$$

$$x = 4$$

4. Ans (3.46)

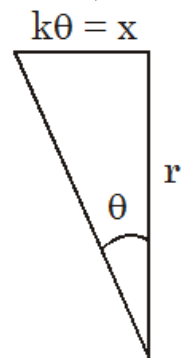


$$\tan \theta = \frac{\frac{d}{2}}{y} = \frac{d}{a}$$

$$\Rightarrow y = \frac{a}{2} = \frac{\sqrt{3}d}{2} = \frac{1.73 \times 4}{2}$$

$$\Rightarrow y = 3.46 \text{ m.}$$

5. Ans (500.00)



(for small angle)

$$F = -kx$$

$$\tau = Fr$$

$$\tau = (kr\theta) r$$

$$I\alpha = (kr^2) \theta$$

$$\alpha = \left(\frac{kr^2}{I} \right) \theta$$

$$T = 2\pi \sqrt{\frac{I}{kr^2}} \quad \dots (i)$$

$$T = 4t_s = 4 \times 31.4 \text{ ms (from graph)}$$

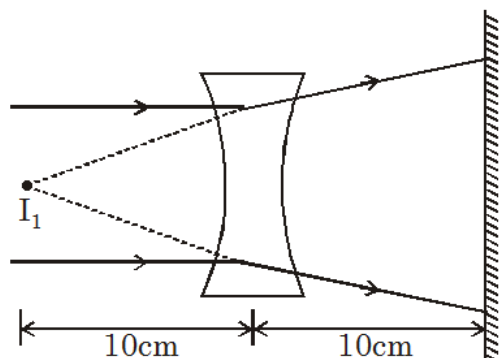
$$T = 40\pi \times 10^{-3} \text{ s} \quad \dots (ii)$$

From (i) and (ii)

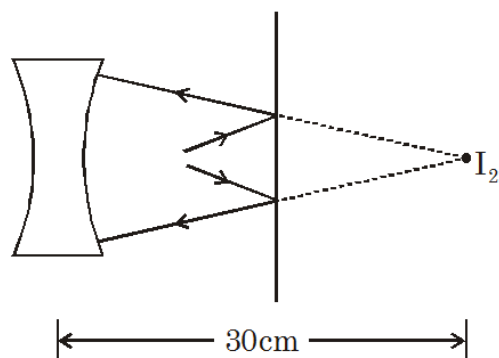
$$I = 500 \times 10^{-6} \text{ kg m}^2/\text{s}^2$$

$$\therefore x = 500$$

6. Ans (2.50)



First image due to lens at focus i.e. 20 cm from mirror.



Final image (I_3) due to lens..

$$\frac{1}{v} - \frac{1}{-30} = \frac{-1}{10} \Rightarrow v = -7.5 \text{ cm}$$

Distance from mirror 2.5 cm

PART-2 : CHEMISTRY

SECTION-I (i)

5. Ans (C)

PART-2 : CHEMISTRY

SECTION-II

2. Ans (53.25)

2g chlorine used 3g metal for formation of metal chloride. Then for chlorine weight 35.5 g weight of metal

$$= \frac{3 \times 35.5}{2} = 53.25$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

$$x^2 - y^2 - 2x + 4y - 7 = 0$$

$$x^2 - 2x + 1 - (y^2 - 4y + 4) = 4$$

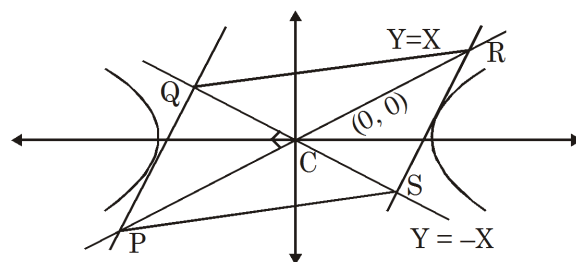
$$(x-1)^2 - (y-2)^2 = 4$$

Perpendicular tangents from the centre are $Y = X$ and $Y = -X$ which are its asymptotes.

$$\therefore \text{Ar}(\Delta PCQ) = ab = 4$$

$$\therefore (\square PQRS) = 16$$

Equation of tangent to the hyperbola with slope = 2.



$$Y = 2X \pm \sqrt{4 \cdot 4 - 4}$$

$$Y = 2X \pm 2\sqrt{3}$$

$$\left. \begin{array}{l} PQ : Y = 2X + 2\sqrt{3} \\ QS : Y = -X \end{array} \right\} \Rightarrow Q \equiv \left(\frac{-2}{\sqrt{3}}, \frac{2}{\sqrt{3}} \right)$$

$$PR : Y = X \Rightarrow P \equiv (-2\sqrt{3}, -2\sqrt{3})$$

$$PQ = \sqrt{24 + \frac{8}{3}}$$

$$= \sqrt{\frac{80}{3}}$$

$$= 4\sqrt{\frac{5}{3}}$$

$$\text{Ar}(\Delta PCQ) = \frac{1}{2} \times \sqrt{24} \cdot \sqrt{\frac{8}{3}} = 4$$

$$\therefore \text{Ar}(\Delta PQR) = 8 \text{ sq.units}$$

$$\text{Ar}(\square PQRS) = 16 \text{ sq.units}$$

2. **Ans (B)**

$$f(x) = \frac{3(x^2 + 1) + ax + b - 3}{1 + x^2}; f(x) = 3 + \frac{ax + b - 3}{1 + x^2}$$

$$y = 3 + \frac{ax + b - 3}{1 + x^2}$$

for y to lie in $[-4, 3]$

$$ax + b - 3 < 0 \quad \forall x \in \mathbb{R}$$

this is possible only if $m = 0$

$$\text{when, } a = 0 \text{ then } y = 3 + \frac{b - 3}{1 + x^2}$$

note that $b - 3 = 0$ then $b < 3$

$$\text{if } x \rightarrow \infty, y_{\max} \rightarrow 3^-$$

now $y_{\min}$ occurs at $x = 0$ (as $1 + x^2$ is minimum,)

$$y_{\min} = 3 + b - 3 = b \Rightarrow b = -4$$

3. **Ans (C)**

$$\text{Given } f(x + 5) \geq f(x) + 5 \quad \dots(1)$$

$$f(x + 1) \leq f(x) + 1 \quad \dots(2)$$

from (1) and (2), we can write

$$\begin{aligned} f(x) + 5 &\leq f(x + 5) \leq f(x + 4) + 1 \\ &\leq f(x + 3) + 2 \\ &\leq f(x + 2) + 3 \\ &\leq f(x + 1) + 4 \\ &\leq f(x) + 5 \end{aligned}$$

$\therefore$ equality exist everywhere

$$\therefore f(x + 1) = f(x) + 1 \quad \because f(1) = 1$$

$$\Rightarrow f(2) = 2, f(3) = 3 \dots f(2021) = 2021$$

$$\text{Now } g(x) = f(x) + 5 - x \Rightarrow g(2021) = 5$$

4. **Ans (C)**

$f(x - 1) = (x^2 - x + 1) \phi(x)$, where $f(x)$ is polynomial.

$$f(x) = (x^2 + x + 1) \phi(x + 1)$$

$$\Rightarrow \phi(x) = \phi(x + 1)$$

$\Rightarrow f(x)$ is a periodic function

$$\phi(x) = \text{constant.}$$

$$f(x) = \lambda(x^2 + x + 1)$$

$$f(x) = (x^2 + x + 1)$$

5. **Ans (C)**

$$([x] + [y] + [z]) (([x] - [y])^2 + ([y] - [z])^2 + ([z] - [x])^2) = 8$$

possible only when $[x] + [y] + [z] = 4$

$$[x] = 2, [y] = 1, [z] = 1$$

$$\text{or } [x] = 1, [y] = 2, [z] = 1$$

$$[x] = 1, [y] = 1, [z] = 2$$

6. **Ans (C)**

$$|a + b + c| = |a| + |b| + |c|$$

when a, b, c has same sign.

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (B,C,D)**

Let $(x_1, y_1) \equiv (at^2, 2at)$ tangent at this points $ty = x + at^2$

$$\text{Any point on this tangent is } \left(h, \frac{h + at^2}{t} \right)$$

Chord of contact of this point w.r.t. the circle $x^2 + y^2 = a^2$ is

$$hx + \left(\frac{h + at^2}{t} \right) y = a^2 \Rightarrow (at y - a^2) + h \left(x + \frac{y}{t} \right)$$

Which is a family of st. lines passing through point of intersection of

$$ty - a = 0 \text{ and } x + \frac{y}{t} = 0$$

$$\text{So the fixed point is } \left(-\frac{a}{t^2}, \frac{a}{t} \right)$$

$$\therefore x_2 = -\frac{a}{t^2}, y_2 = \frac{a}{t}$$

$$\therefore x_1 x_2 = -a^2, y_1 y_2 = 2a^2$$

8. **Ans (B,D)**

$y = m(x - a)$ is a focal chord so, angle between normals drawn at A and B = 90°

$$\text{length of focal chord} = 4a \operatorname{cosec}^2 \alpha$$

$$\text{for } m = 1, \alpha = \frac{\pi}{4} \Rightarrow \operatorname{cosec}^2 \frac{\pi}{4} = 2$$

$$AB = 4a \times 2 = 8a$$

9. Ans (A,C)

A cyclic parallelogram will be a rectangle or square So, $\angle QPR = 90^\circ$

$\Rightarrow$ P lies on director circle of ellipse

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$$

i.e. $x^2 + y^2 = 25$ is director circle of

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow 16 + b^2 = 25 \Rightarrow b^2 = 9 \Rightarrow a^2(1 - e^2) = 9$$

$$\Rightarrow 1 - e^2 = \frac{9}{16} \Rightarrow e^2 = \frac{7}{16} \Rightarrow e = \frac{\sqrt{7}}{4}$$

10. Ans (A,C,D)

Since $49^{x^2+y} + 49^{y^2+x}$ are both positive we can apply

AM-GM to obtain

$$\frac{1}{\sqrt{7}} = \frac{49^{x^2+y} + 49^{y^2+x}}{2} \geq \sqrt{49^{x^2+y+y^2+x}}$$

$$\frac{1}{\sqrt{7}} = \frac{49^{x^2+y} + 49^{y^2+x}}{2} \geq 7^{x^2+y+y^2+x} = 7^{\left(x+\frac{1}{2}\right)^2 + \left(y+\frac{1}{2}\right)^2}$$

$$\text{However, } 7^{\left(x+\frac{1}{2}\right)^2 + \left(y+\frac{1}{2}\right)^2} \geq 7^0 \geq 1$$

Hence we must have

$$\left(x + \frac{1}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = 0 \text{ this is true for only}$$

$$x = y = -\frac{1}{2} \text{ only.}$$

11. Ans (A,C,D)

$$8^3x^9 + 9.8^2x^6 + 27.8x^3 - 512x + 219 = 0$$

$$(8x^3)^3 + 3.(8x^3)^2.3 + 3(8x^3).(3)^2 + 27 = 512x - 192$$

$$[8x^3 + 3]^3 = 512x - 192$$

$$8x^3 + 3 = (512x - 192)^{1/3}$$

$$x^3 + \frac{3}{8} = \left(x - \frac{3}{8}\right)^{1/3}$$

$$\Rightarrow f(x) = f^{-1}(x)$$

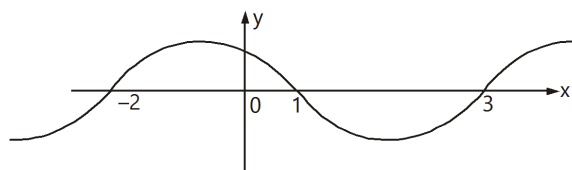
$$\Rightarrow f(x) = x [\because f(x) \text{ is continuous and } \uparrow \text{ function}]$$

$$\Rightarrow x^3 + \frac{3}{8} = x$$

$$\Rightarrow (2x-1)(4x^2+2x-3) = 0$$

$$\Rightarrow x = \frac{1}{2}, \frac{-1+\sqrt{13}}{4}, \frac{-1-\sqrt{13}}{4}$$

12. Ans (A,B)



$$y = f(x)$$

$$(A) |y| = \text{sgn } f(x)$$

$$f(x) > 0 \text{ if } -2 < x < 1 \text{ and } x > 3$$

$$\Rightarrow |y| = 1$$

$$f(x) < 0, 1 < x < 3$$

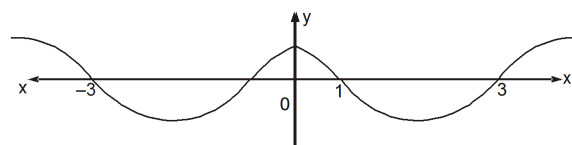
$$\Rightarrow |y| = -1 \text{ (not applicable)}$$

$$\text{at } x = -2, 1, 3 \quad f(x) = 0$$

$$\Rightarrow |y| = 0$$

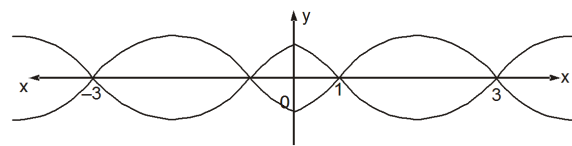
$$(B) |y| = \text{sgn } (-f(|x|))$$

$$f(|x|) =$$



$$(C) |y| = |f(|x|)|$$

$$|f(|x|)| =$$



$$(D) y = x^{\text{sgn}(x)}$$

$$f(x) > 0, -2 < x < 1 \Rightarrow y = x$$

$$f(x) < 0, 1 < x < 3 \Rightarrow y = \frac{1}{x}$$

$$x = -2, 1, 3 \Rightarrow y = 1$$

$$x > 3, f(x) > 0 \Rightarrow y = x$$

PART-3 : MATHEMATICS

SECTION-II

2. **Ans (1.50)**

$$\frac{T_{n+1}}{2^{n+1}} = \frac{T_n}{2^n} + \frac{1}{2} \times 2^n$$

$$\Rightarrow \frac{T_2}{2^2} - \frac{T_1}{2} = \frac{1}{2} \times 2$$

$$\frac{T_3}{2^3} - \frac{T_2}{2^2} = \frac{1}{2} \times 2^2$$

$$\frac{T_n}{2^n} - \frac{T_{n-1}}{2^{n-1}} = \frac{1}{2} \times 2^{n-1}$$

$$\frac{T_n}{2^n} - \frac{T_1}{2} = \frac{1}{2} \frac{(2^{n-1} - 1) \times 2}{2^{-1}} = 2^{n-1} - 1$$

$$\Rightarrow T_n = 2^{2n-1} - 2^{n-1}$$

$$\Rightarrow S_n = \frac{2 \cdot 4^n - 2 - 3 \cdot 2^n + 3}{3}$$

$$= \frac{2 \cdot 4^n - 3 \cdot 2^n + 1}{3}$$

3. **Ans (1.00)**

$$\frac{\log a}{\log b} \cdot \frac{\log a}{\log c} + \frac{\log b}{\log a} \cdot \frac{\log b}{\log c} + \frac{\log c}{\log a} \cdot \frac{\log c}{\log b}$$

$$= \frac{\log a \sqrt{a}}{\log b} \times \frac{\log b^{1/3}}{\log \sqrt{c}} \times \frac{\log c^3}{\log a} = 3$$

$$\Rightarrow \log^3 a + \log^3 b + \log^3 c = 3 \log a \cdot \log b \cdot \log c$$

$$\Rightarrow \log a + \log b + \log c = 0 \Rightarrow abc = 1$$

4. **Ans (256.00)**

Periodic of individual functions are $2\pi, 2\pi, 2^2\pi,$

$2^3\pi, \dots, 2^8\pi$ & their L.C.M is $2^8\pi$

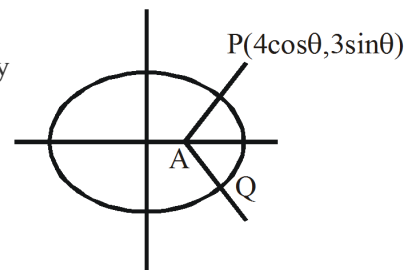
5. **Ans (3.24)**

if AP & AQ perpendicular

$$\Rightarrow m_{AP} = 1$$

$$4x \sec \theta - 3y$$

$$\operatorname{cosec} \theta = 7$$



$$m = \frac{4}{3} \tan \theta = 1$$

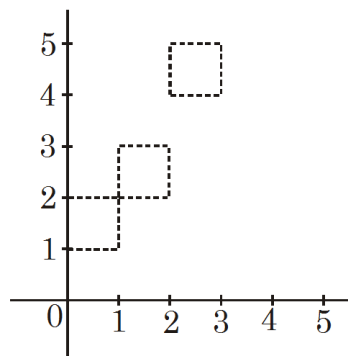
$$\tan \theta = \frac{3}{4}$$

$$\Rightarrow P \left(\frac{16}{5}, \frac{9}{5} \right); Q \left(\frac{16}{5}, -\frac{9}{5} \right)$$

$$A \left(\frac{7}{5}, 0 \right)$$

$$\text{Area} = \frac{1}{2} \cdot \frac{18}{5} \cdot \frac{9}{5} = \frac{81}{25} = 3.24$$

6. **Ans (9.00)**



$$\text{If } x \in [0, 1) \Rightarrow y \in [1, 2)$$

$$\text{If } x \in [1, 2) \Rightarrow y \in [2, 3)$$

$$\text{If } x \in [2, 3) \Rightarrow y \in [3, 4)$$

$$\text{If } x \in [3, 4) \Rightarrow y \in [4, 5)$$

$$\dots$$

$$\text{If } x \in [8, 9) \Rightarrow y \in [256, 257)$$

Hence,

$$\text{total area} = 1 + 1 + 1 \dots = 9$$